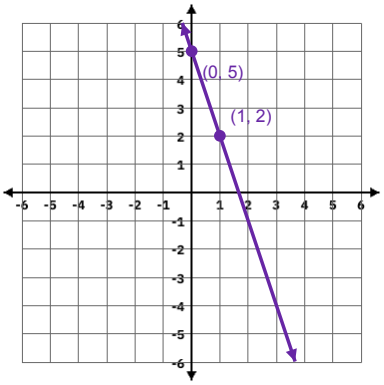
PRACTICE UNIT 1B Lesson 3 U1B L3 Graphing Linear Equations Name: _____HR:_____

Directions: Graph accurately showing two points on every graph, fill in the blanks.

WORKED KEY

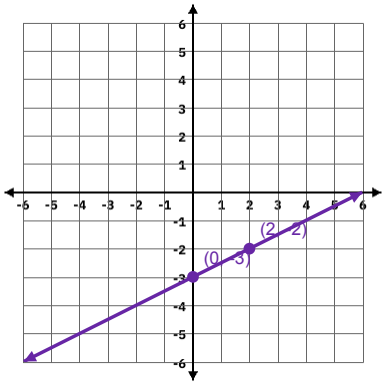
1. $f(x) = -3x + 5$

m = **-3** b = **5**



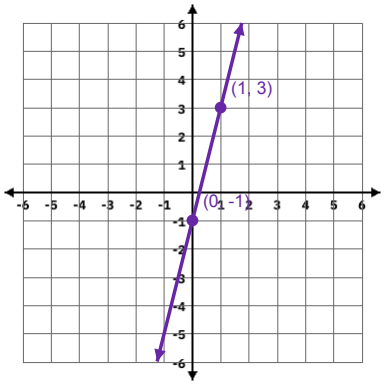
2. $g(x) = 1/2x - 3$

m = $\frac{1}{2}$ b = **-3**



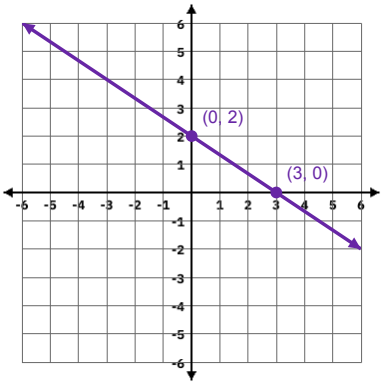
3. $h(x) = 4x - 1$

m = **4** b = **-1**



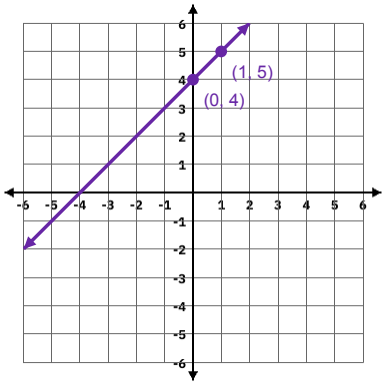
4. $p(x) = \frac{-2}{3}x + 2$

m = $\frac{-2}{3}$ b = **2**



5. $y = x + 4$

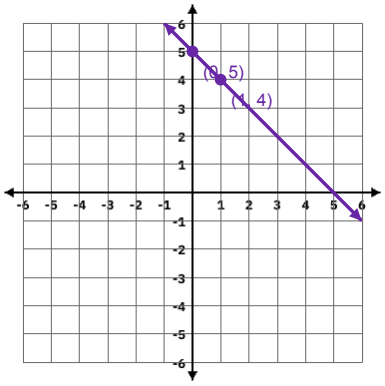
m = **1** b = **4**



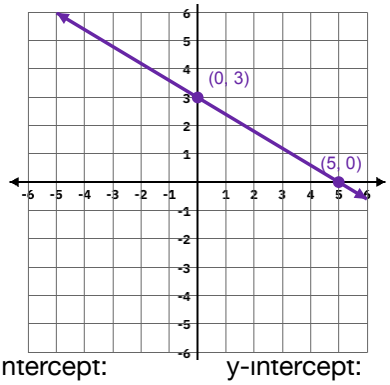
6. $y = 5 - x$ (put in slope-int form)

Slope-intercept form: **y = −x + 5**

m = **-1** b = **5**



7. $3x + 5y = 15$ FIND x and y intercepts. Then graph.

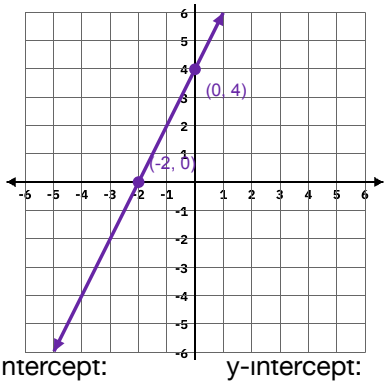


x-intercept: **(5, 0)** y-intercept: **(0, 3)**

y = 0: 3x = 15, x = 5

x = 0: 5y = 15, y = 3

8. $-2x + y = 4$ FIND x and y intercepts. Then graph.

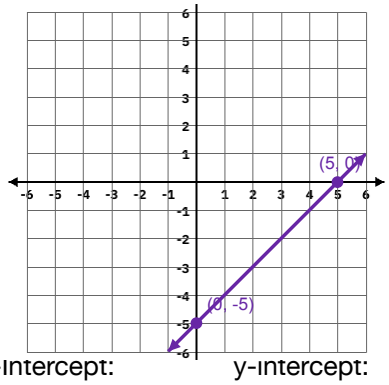


x-intercept: **(-2, 0)** y-intercept: **(0, 4)**

y = 0: −2x = 4, x = −2

x = 0: y = 4

9. $7x - 7y = 35$ FIND x and y intercepts. Then graph.



x-intercept: **(5, 0)** y-intercept: **(0, -5)**

y = 0: 7x = 35, x = 5

x = 0: −7y = 35, y = −5

U1B L3 Graphing Linear Equations Name: _____HR:_____

Directions: Put in SLOPE-INTERCEPT form and graph.

WORKED KEY

10. $4y - 8x = 12$

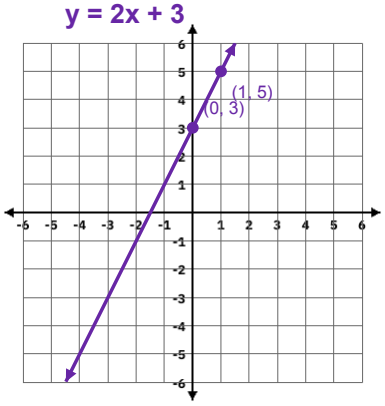
$4y - 8x = 12$

$+8x \quad +8x$

$4y = 8x + 12$

$\frac{4y}{4} = \frac{8x + 12}{4}$

y = 2x + 3



11. $2x - y = 6$

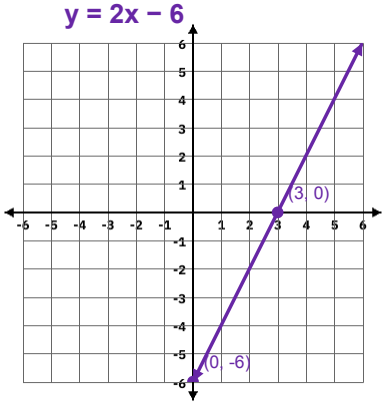
$2x - y = 6$

$-2x \quad -2x$

$-y = -2x + 6$

$\frac{-y}{-1} = \frac{-2x + 6}{-1}$

y = 2x − 6



12. $9x - 3y = 18$

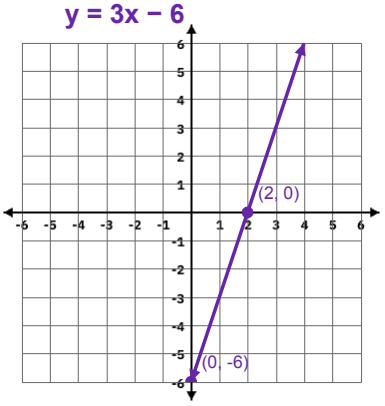
$9x - 3y = 18$

$-9x \quad -9x$

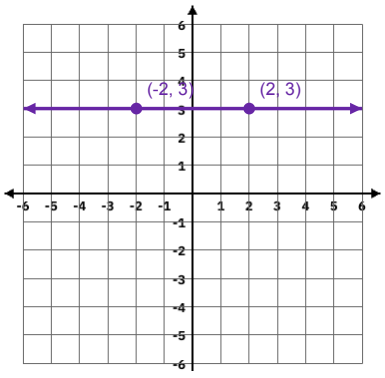
$-3y = -9x + 18$

$\frac{-3y}{-3} = \frac{-9x + 18}{-3}$

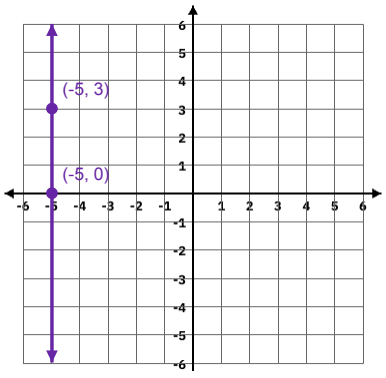
y = 3x − 6



Horizontal lines have a slope m = **0**. Graph the line $y = 3$. Try it on desmos.



Vertical lines have a slope = m = **undefined**. (This means there is not a number that describes the lines steepness, the line is just called vertical and everyone then knows it goes straight up!) Graph: $x = -5$



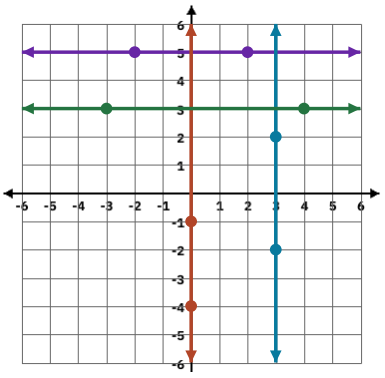
13. Graph the following: (Get the variable alone if needed.)

a. $y = 5$ **y = 5**

b. $2x = 6$ **x = 3**

c. $x + 3 = 3$ **x = 0**

d. $y - 3 = 0$ **y = 3**



14. A fishing line is 120 inches long when no weight is attached. When a weight is attached, the line stretches an additional 0.35 inches for every pound of weight.

Write a function that models the total length of the fishing line as a function of the weight attached.

L(w) = 0.35w + 120

w: weight in pounds (w ≥ 0)

L(w): total length in inches

m = 0.35 inch per pound

15. You have $50 in your wallet and earn $15 per hour babysitting.

a. Write a function $M(t)$ in slope-intercept form.

M(t) = 15t + 50

b. How many hours do you need to work to save $140? (Show work)

$15t + 50 = 140$

$-50 \quad -50$

$15t = 90$

$t = 6$ hours

16. A tank initially contains 200 liters of water and drains at a constant rate of 8 liters per minute.

a. Let $V(t)$ be the volume in liters after t minutes.

t is time in minutes.

b. Write $V(t)$ in slope-intercept form.

V(t) = −8t + 200

c. How long does it take to empty the tank? (Show work)

$0 = -8t + 200$

$-200 \quad -200$

$-200 = -8t$

$t = 25$ minutes